

### 5.1 Quiz: Maclaurin Series (no calculator)

Answer all three problems. Show full working in the spaces provided. Total: 28 marks. Time: 45 minutes.

1. [Maximum mark: 7]

AA HL 2025 Nov P1 TZ1 Q05

The first four terms of the Maclaurin series expansion of  $(1 - x)^{-4}$  are

$$1 + ax + bx^2 + 20x^3,$$

where  $a, b \in \mathbb{Z}^+$ .

- (a) (i) Show that  $a = 4$ . [2]
- (ii) Find the value of  $b$ . [3]
- (b) A car was purchased four years ago. The car depreciated in value by 10% each year. The value of the car today is \$1000. Using the results of part (a), estimate the value of the car four years ago. [2]

2. [Maximum mark: 7]

AA HL 2024 May P1 TZ1 Q08

- (a) Find the first two non-zero terms in the Maclaurin series of

$$(i) \sin(x^2); \quad [2]$$

$$\text{(ii) } \sin^2(x^2). \quad [3]$$

- (b) Hence, or otherwise, find the first two non-zero terms in the Maclaurin series of  $4x \sin(x^2) \cos(x^2)$ . [2]

3. [Maximum mark: 14]

AA HL 2021 Nov P1 TZ0 Q11

(a) Prove by mathematical induction that

$$\frac{d^n}{dx^n}(x^2 e^x) = [x^2 + 2nx + n(n-1)] e^x \quad \text{for } n \in \mathbb{Z}^+.$$

[7]

- (b) Hence or otherwise, determine the Maclaurin series of  $f(x) = x^2e^x$  in ascending powers of  $x$ , up to and including the term in  $x^4$ . [3]

- (c) Hence or otherwise, determine the value of

$$\lim_{x \rightarrow 0} \left[ \frac{(x^2e^x - x^2)^3}{x^9} \right].$$

[4]